NIHVA
NIHVA Technologies Pvt. Ltd.

SYSTEM & MAINTENANCE MANUAL

FOR

SEMI ELECTRIC DC SCISSOR LIFT

CLIENT: TENNECO AUTOMOTIVE INDIA PVT LTD.

MFG.YEAR: - JUNE 2020.

DESIGNED, DEVELOPED & MARKETED

BY



G-BLOCK, 1ST FLOOR, THERMAX CHOWK
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SCISSOR LIFT	
SIZE (L X W)	1000 mm X 800 mm
CAPACITY	500 KG
SR NO	NSCL579
CLIENT	TENNECO AUTOMOTIVE INDIA PVT LTD.

1. GENERAL INFORMATION

1. The hydraulic scissor lift load capacity rating and the serial number is mentioned on metal name plate.
2. Lift should not be overloaded beyond the rated load capacity may result in damage.
3. The hydraulic system consists of pump, motor, oil tank & valves.
4. The gear pump is set at operating pressure of 200 bar.
5. The operating principle is that the pump is operated to raise the table, and the pump is stopped when the table attains the desired elevation. A check valve and lowering valve between the pump and cylinder holds the table at elevation. For lowering to any desired level, the solenoid valve is energized to allow fluid to return from hydraulic cylinder to reservoir.
6. A scissor lift should only be operated on a solid surface such as concrete, tile, or a compacted road base. When lifts are placed on uneven surfaces or earth, mulch or gravel that is not compacted, the lifts may tip and fall over, causing serious injury or death to the people on the platform.

1.1 MAIN SPECIFICATIONS

A) Maximum Load Lifting Capacity	500 KG
B) Max Height of Platform	800 MM
C) Actual Lift	500 MM
D) Min Height of Platform	300 MM
E) Platform Length	1000 MM
F) Platform Width	800 MM
G) Power Pack	1.6kw, 12Volt
H) Oil Tank Capacity	5 LTR

2. ELECTRO HYDRAULIC SCISSOR LIFT- DO's & DONT's: -

DOs: -

- Check all the electrical connections.
- Check the hydraulic connections.
- Ensure NO Oil Leakage from joints.
- Ensure the oil level in the oil tank before using the lift.
- Ensure that there is not any obstacle within the region of moving parts.
- While doing maintenance work, ensure that the Emergency switch is pressed & the maintenance stoppers are in their position while working below the lift.

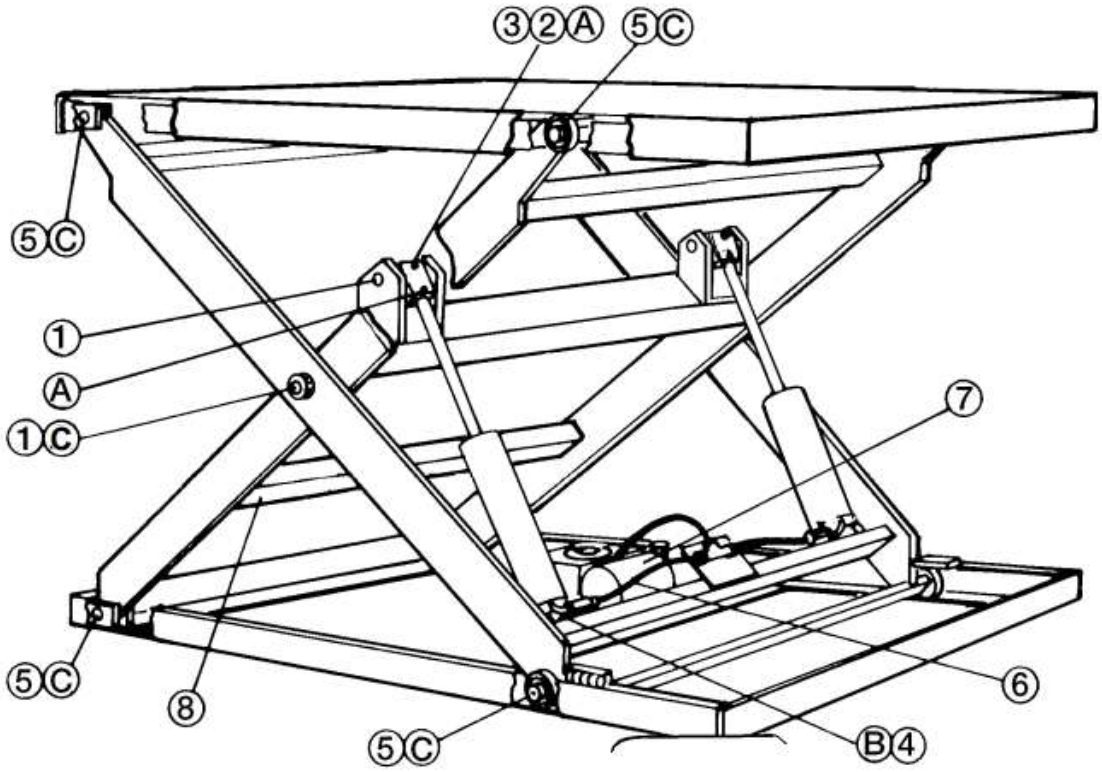
DON'Ts: -

- Lift should not be loaded beyond its rated load capacity.
- Don't loose the oil hoses without resting the lift on its maintenance stoppers.
- Do not operate damaged lift.
- Don't stand at the edges of lift platform while the lift is lowering.
- Don't change lowering valve setting without supervision or any assistance.
- In case of power failure, to resume the downward operation after Power ON, do not lower the lift. Instead, take the lift slightly upward & then resume to lower it.
- Don't put any manual stopper or anything in the roller track.

3. OPERATING INSTRUCTIONS

a) Check all electrical connection for short or phase short. Check hydraulic connections are tightened properly to ensure no oil leakage from joints.
b) Connect (1 phase 230V) supply to the panel through cable top for battery charging.
c) Switch Emergency push button is in unlock position.
d) Use Lift's PUSH BUTTONS for moving lift upward & downward.
e) In case of emergency, press emergency STOP push button provided on push button station or pendent.
f) For controlling the lowering speed, adjust the valve provided on manifold.

4. LUBRICATION & ROUTINE MAINTENANCE INSTRUCTIONS



- **Lubrication Points**

Lubrication points as mentioned below,

A. Points noted with ‘A’, apply grease lubrication.

B. Points noted with ‘B’, apply oil lubrication.

C. Points noted with ‘C’, no lubrication required (Pre-Lubricated &sealed)

Ensure all Moving parts like Wheel, Flap Joint, Rollers etc. are greased.

- **Inspection Check Points**

a) Visually inspect all jack pins, pivot pins to ensure that the set screw has not loosened, allowing pins to move.

b) Confirm Hex Nut Bolt, Allen Bolt -Pivot Pin Locking are Fixed firmly.

c) Confirm Grub screw, Cup Washer on Scissor Arm Joint Bolt are fixed & are tightened firmly.

d) Confirm Tightening of Rollers & its free movement.

e) Confirm all Bolts at Scissor Arm are tightened.

f) Visually inspect all lift arm rolling bearings to ensure that bearing bolts are tight.

g) Visually inspect all hydraulic hoses and fittings to ensure there are no cut on hoses or leaking fittings.

h) Visually inspect all electrical lines to ensure there are no cut wires or loose cords.

i) Visually inspect the entire lift structure.

j) These recommendations are applicable to the equipment considering daily usage of approx. 10 to 15 times.

If usage is more frequent, the same time intervals for maintenance should be adjusted accordingly.

- **WEEKLY**

a) Do greasing / oiling of all moving joints.

- **MONTHLY**

a) Check hydraulic pipeline for leak proof working.

b) Check oil level in hydraulic tank.

c) Check cylinder seals & ‘O` rings for leak proof working.

- **YEARLY**

a) Change hydraulic oil.

b) Check main shaft seal & shaft ‘O` ring for leak proof working.

Hydraulic oil to be used – Enclo – 68 OR Servo – 68.

5. GUIDE LINES IN CASE OF OIL LEAKAGE

a) Leakage at Piping –

- Tighten Joints.
- Replace Dowty Seal at Mounting

b) Leakage in Cylinder-

- Replace Seals.

c) Replace Oil after Every Year-

- Drain the Oil from the System.
- Clean filter, Tank elements, and strainers before refilling.

6. TROUBLE SHOOTING GUIDELINES

i. If lift will not raise:

a. Check electric circuit on new installations of motors, if lift does not start raising in approximately 10 seconds.
b. Check oil level. With lift in full up position, oil level should be Approximately 1 1⁄4" above tank bottom.

ii. ***Lift raises slower than specified rate or will raise only partial load:***

Note: Allowance must be maintained for slight speed variations (due to temperature which changes the viscosity of the hydraulic oil varies)
a. Check line voltage under load condition. Low voltage affects capacity of equipment.
b. If foam is visible in tank oil, check for a loose connection in suction line between pump and tank. Tighten connections.

iii. Lift slowly settles with lowering valve not energized:

a. Flush lowering valve by cycling lift up and down. This should be done with no load.
b. If lift continues to settle, replace down valve cartridge.
c. If lift continues to settle, replace pump.

iv. If Lift will not lower:

a. Look for mechanical restrictions.
b. Test valve coil for operation. Check voltage at coil.
c. If voltage is ok, replace coil.

v. If lift will not raise to full vertical travel:

a. Check for low oil level if necessary, add oil.

vi. If lift does not lower smoothly:

a. Run the lift up and down under load condition for some time to purge air.

7. GENERAL SAFETY INSTRUCTIONS

Use trained personnel for loading & unloading of equipment. The equipment must be properly handled when transferring from the unloading area to final site to prevent damage.

DO NOT operate equipment unless it is properly locked. It may cause serious injury or death.

a) Install all safety devices, guards and guarding prior to equipment start-up.
b) Only Trained & Authorized person must use the lift.
c) Always use mechanical stopper if you are going to work in elevated condition.
d) Keep working area free from oil to Avoid Slipping/Fire hazards.
e) Do Not Exceed the Lift Capacity than rated.
f) Before cleaning or servicing disconnect power supply
g) As the lift moves up and down, "pinch points" are formed. Keep hands, feet, hair, loose clothing, etc. away from these pinch points. Any of these may cause serious personal injury or death.
h) Ensure that the hydraulic and electrical lines will not be pinched by the lift as it rises or lowers. If a line gets pinched, the lift may not work properly. A hose may break, the lift table may drop suddenly, and someone may get injured. If an electrical line is broken, may cause personal injury or death due to electrical shock.

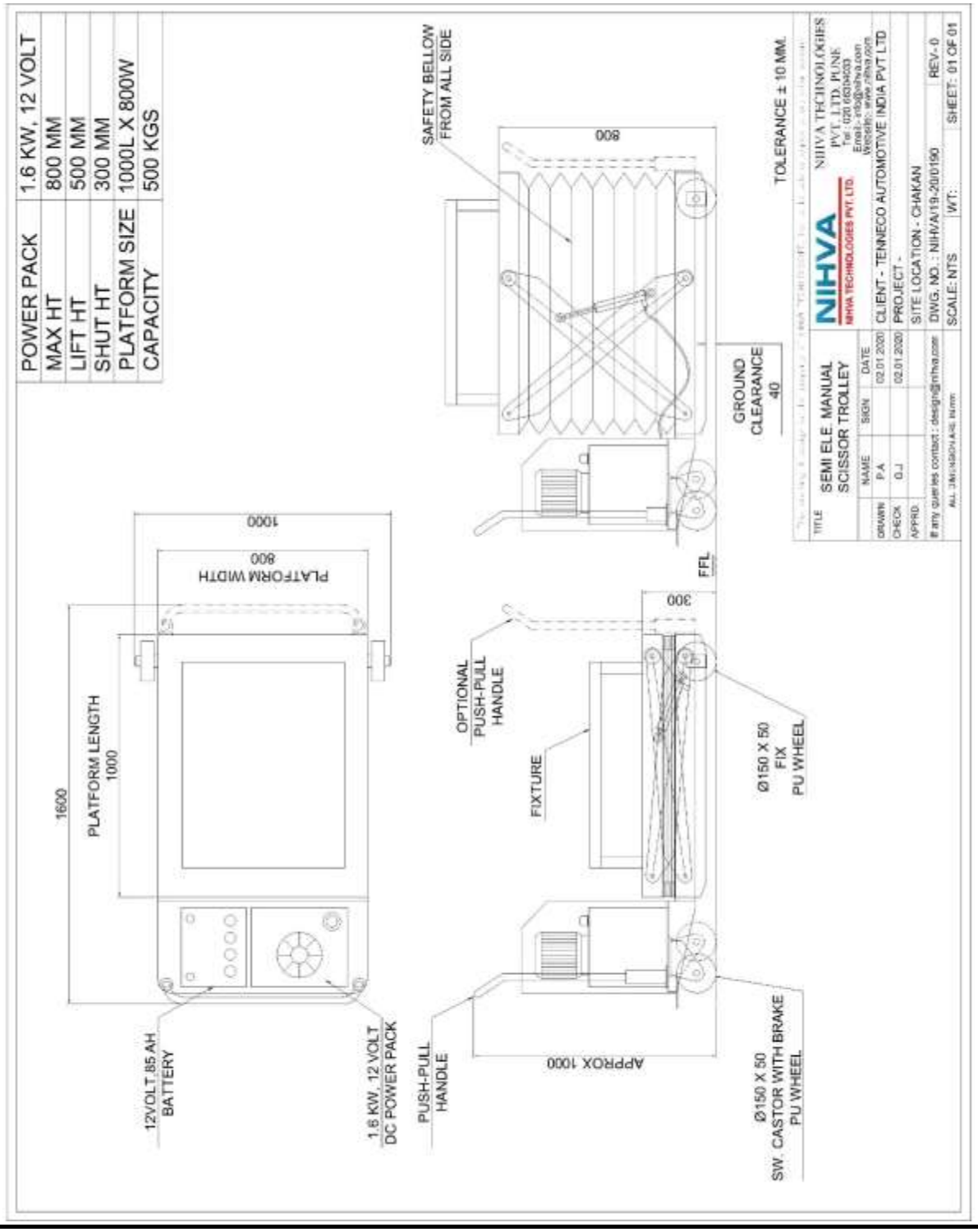
i) Read the motor nameplate and record the motor full load current. Consult local electrical codes for power supply fusing requirements. To avoid fire hazards, follow these fusing requirements.



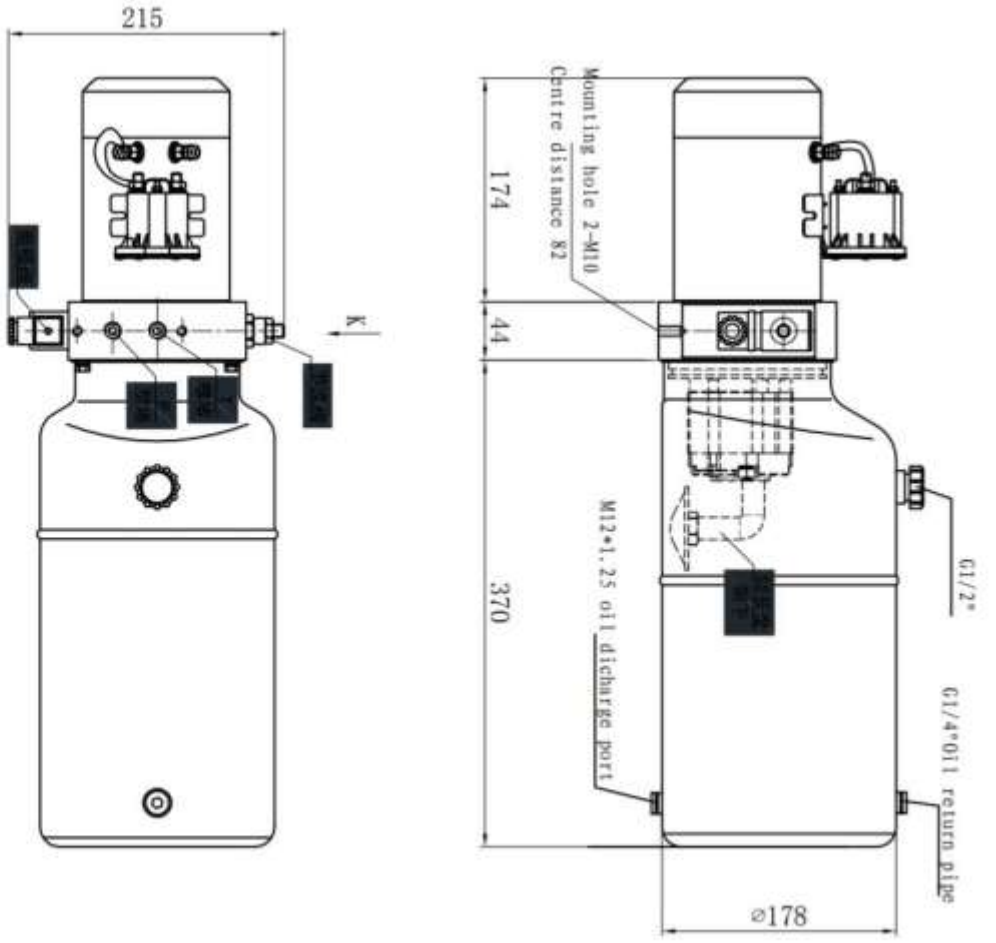




8. GA DRAWING.


POWER PACK 1.6 KW, 12 VOLT
MAX HT 800 MM
LIFT HT 500 MM
SHUT HT 300 MM
PLATFORM SIZE 1000L X 800W
CAPACITY 500 KGS
SAFETY BELLOW FROM ALL SIDE
TOLERANCE ± 10 MM.
GROUND CLEARANCE 40
FFL
PLATFORM WIDTH
PLATFORM LENGTH
OPTIONAL PUSH-PULL HANDLE
FIXTURE
Ø150 X 50 FIX PU WHEEL
12V/LT.85 AH BATTERY
1.6 KW, 12 VOLT DC POWER PACK
PUSH-PULL HANDLE
Ø150 X 50 SW. CASTOR WITH BRAKE PU WHEEL
SEMI ELE. MANUAL SCISSOR TROLLEY
NIHVA TECHNOLOGIES PVT. LTD.
CLIENT - TENNECO AUTOMOTIVE INDIA PVT LTD
PROJECT -
SITE LOCATION - CHAKAN
DWG. NO. : NIHVA/19-20/0190
REV- 0
SCALE: NTS
WT:
SHEET: 01 OF 01

9. HYDRAULIC POWER PACK



10. ELECTRICAL CIRCUIT: -

